



**SCHOOL OF
ENGINEERING**

DAYANANDA SAGAR UNIVERSITY

School of Engineering

Department of Electronics and Communication Engineering Bengaluru – 561112

M.Tech Thesis

“FPGA-Based Design and Implementation of a Reconfigurable Multi-Modulation Signal Generator Using Direct Digital Synthesis”

Submitted by

Supreeth K

USN: ENG24EMS0003

Under the Guidance of

Dr. Shirshendu Roy

Professor, Department of ECE

Submitted in partial fulfillment of the requirements for the degree of

Master of Technology

in

Embedded Systems

2026



SCHOOL OF
ENGINEERING

DAYANANDA SAGAR UNIVERSITY

School of Engineering

Department of Electronics and Communication Engineering

Bengaluru – 561112

SUPERVISOR CERTIFICATE

This is to certify that the thesis entitled “**FPGA-Based Design and Implementation of a Re-configurable Multi-Modulation Signal Generator Using Direct Digital Synthesis**”
is a Bonafide record of the project work carried out by

Supreeth K (USN: ENG24EMS0003)

in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Embedded Systems** at **Dayananda Sagar University, School of Engineering, Bengaluru.**

This work has been carried out during the academic year **2025–2026** under the guidance of **Dr. Shirshendu Roy**, Professor, Department of Electronics and Communication Engineering.

To the best of my knowledge, this thesis has not been submitted elsewhere for the award of any other degree or diploma.

Dr. Shirshendu Roy

Professor, Department of ECE

Dayananda Sagar University

Signature:

Chairman of the Department

Department of Electronics and Communication Engineering Dayananda Sagar University

Signature:

Date:

Place: Bengaluru



**SCHOOL OF
ENGINEERING**

DAYANANDA SAGAR UNIVERSITY

School of Engineering

Department of Electronics and Communication Engineering

Bengaluru – 561112

Certificate of Examination

This is to certify that the thesis entitled “**FPGA-Based Design and Implementation of a Re-configurable Multi-Modulation Signal Generator Using Direct Digital Synthesis**” submitted by

Supreeth K (USN: ENG24EMS0003)

in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Embedded Systems** at **Dayananda Sagar University, School of Engineering, Bengaluru** has been examined and approved by the undersigned examiners.

External Examiner

Name:

Signature:

Supervisor

Dr. Shirshendu Roy

Department of ECE

Signature:

Chairman of the Department

Department of Electronics and Communication Engineering Dayananda Sagar University

Signature:

Date:

Place: Bengaluru



SCHOOL OF
ENGINEERING

DAYANANDA SAGAR UNIVERSITY

School of Engineering

Department of Electronics and Communication Engineering

Bengaluru – 561112

Declaration

I hereby declare that the thesis entitled “**FPGA-Based Design and Implementation of a Re-configurable Multi-Modulation Signal Generator Using Direct Digital Synthesis**” is a Bonafide record of the work carried out by us in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Embedded Systems** at **Dayananda Sagar University, School of Engineering, Bengaluru**.

This work has been carried out under the guidance of **Dr. Shirshendu Roy**, Department of Electronics and Communication Engineering.

I further declare that this work has not been submitted previously, either in full or in part, to any other university or institution for the award of any degree or diploma.

Supreeth K

USN: ENG24EMS0003

Signature:

Date:

Place: Bengaluru



SCHOOL OF
ENGINEERING

DAYANANDA SAGAR UNIVERSITY

School of Engineering
Department of Electronics and Communication Engineering
Bengaluru – 561112

Acknowledgement

I express my sincere gratitude to Dayananda Sagar University, School of Engineering, Bengaluru for providing me with the opportunity to carry out this thesis work.

I would like to express my heartfelt thanks to my guide **Dr. Shirshendu Roy.**, Assistant Professor, Department of Electronics and Communication Engineering, for his invaluable guidance, continuous support, and encouragement throughout the course of this work.

I also extend my sincere thanks **to Dr. Arun Balodi**, Professor and Chairman, Department of Electronics and Communication Engineering, for his valuable suggestions and support.

I would like to thank the faculty members and staff of the Department of Electronics and Communication Engineering for their assistance and support.

Finally, I express my gratitude to my family and friends for their constant encouragement and support.

Supreeth K

Abstract

This thesis presents the design and implementation of a reconfigurable multi-modulation signal generator using FPGA technology and Direct Digital Synthesis (DDS). In modern communication and radar systems, the ability to generate different types of modulated signals with high accuracy and flexibility is essential. However, conventional signal generators are often expensive, less adaptable, and limited in functionality.

To overcome these limitations, this work focuses on developing a single FPGA-based platform capable of generating multiple modulation schemes, specifically Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM), and Linear Frequency Modulation (LFM). The system is designed using Verilog HDL, where DDS is used to generate precise carrier and modulating signals with controllable frequency, phase, and amplitude. The modular design approach allows switching between different modulation techniques without requiring hardware changes.

The proposed design is first verified through simulation to analyze waveform behavior and confirm correct modulation characteristics. Following this, the design is implemented on FPGA hardware, and the generated signals are observed using a Cathode Ray Oscilloscope (CRO). The experimental results validate the successful generation of the implemented modulation schemes and show good agreement with simulation results, demonstrating the correctness and reliability of the proposed system.

The developed system provides a cost-effective, flexible, and scalable solution for multi-modulation signal generation. It is suitable for educational laboratories, testing environments, and research applications, and can be extended further to support advanced communication techniques and real-time control interfaces.

List of Figures

2.1 Block Diagram of Amplitude modulation.....	9
2.2 Block Diagram of Frequency modulation.....	10
2.3 Block Diagram of Phase modulation.....	10
2.4 Block Diagram of Linear Frequency modulation.....	12
2.5 Block Diagram of Direct Digital Synthesis.....	13
3.1 Optimized Multi-Modulation Signal Generator Architecture.....	17
3.2 Design Flow of the Proposed Multi-Modulation Signal Generator.....	19
3.3 MATLAB Generated AM, FM, PM, and LFM Signals.....	20
3.4 Simulation Results of (a) AM, (b) FM, (c) PM, and (d) LFM Signals.....	23
3.5 Experimental Setup of Multi-Modulation Signal Generation.....	24
3.6 Simulation and Hardware Verification Summary.....	27

List of Tables

2.1 Comparison of Modulation Techniques.....	14
3.1 Simulation and Hardware Verification Summary.....	27

Contents

List of Figures	vi
Lists of Tables	vii
1 Introduction	1
1.1 Motivation.....	1
1.2 Scope of Research.....	1
1.3 Objectives.....	2
1.4 Thesis Organization.....	3
1.5 Summary.....	3
2 Theoretical Background	4
2.1 Introduction.....	4
2.2 Previous Work.....	5
2.3 Research Gap.....	7
2.4 Modulation Techniques.....	8
2.5 Summary.....	14
3 Proposed Work and Methodology	15
3.1 Introduction.....	15
3.2 Multi-Modulation Signal generation.....	16
3.3 Overall System Architecture.....	17
3.4 Design Methodology.....	18
3.5 Results.....	20

3.6 Discussion.....	27
4 Conclusion and Future Scope	29
4.1 Conclusion.....	29
4.2 Future Scope.....	30
Abbreviations	31
References	32
Dissemination of Thesis	35

Chapter 1

Introduction

1.1 Motivation

Signal generation is a fundamental requirement in modern electronics and communication systems, with applications ranging from wireless communication and radar systems to testing and academic laboratories. Accurate and controllable signal generation is essential for analyzing system performance and validating communication techniques. However, conventional signal generators are often limited in flexibility, as they are designed for specific functionalities and cannot easily support multiple modulation schemes.

In addition, advanced commercial signal generators are expensive, making them less accessible for academic and research purposes. The need for separate hardware to generate different modulation techniques such as Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM) and Linear Frequency Modulation (LFM) increases the complexity and cost of the system.

With the rapid advancement of digital communication systems, there is a growing demand for flexible, reconfigurable, and cost-effective signal generation platforms. FPGA technology provides a suitable solution by enabling multiple signal generation techniques to be implemented on a single hardware platform. Its ability to support high-speed operation and precise timing control makes it ideal for modern signal processing applications. Therefore, the motivation of this work is to design and implement a reconfigurable FPGA-based signal generator capable of generating multiple modulation schemes using Direct Digital Synthesis (DDS). This approach aims to provide a flexible, accurate, and economical solution for signal generation in academic and research environments.

1.2 Scope of Research

The design and implementation of FPGA based multi-modulation reconfigurable signal generator is studied in this work. The system is designed to produce different modulation schemes such as Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM), and Linear Frequency Modulation (LFM) which are extensively used in communication and radar

applications. The aim is to create a single platform capable of generating a number of signals with precise control over key parameters.

The proposed system is based on Direct Digital Synthesis (DDS) which allows accurate and stable generation of signals by digital control of the frequency, phase and amplitude. The design is implemented using Verilog HDL and developed by using Xilinx Vivado tools. A modular architecture is used, with separate functional blocks for waveform generation, modulation and control. This approach makes the system much more flexible and allows us to easily modify or extend it whenever needed.

The performance of the system was verified through simulation as well as hardware implementation. Simulation studies were initially conducted to analyze the waveforms and validate the applied modulation techniques. After successful simulation, the design is implemented on an FPGA board, and the output signals are observed using a Cathode Ray Oscilloscope (CRO) to confirm correct operation. The scope of this work is limited to signal generation and verification. It does not include transmission, receiver design, or wireless communication implementation. Advanced modulation schemes and real-time user interface control are also beyond the scope of this project but can be considered for future enhancements.

1.3 Objectives

- To design and implement a reconfigurable signal generator using FPGA technology
- To produce different modulation schemes like Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM) and Linear Frequency Modulation (LFM)
- To utilize Direct Digital Synthesis (DDS) for precise control of frequency, phase, and amplitude
- To develop the system using Verilog HDL and implement it using Xilinx Vivado tools
- To verify the functionality of the design through simulation
- To implement the design on FPGA hardware and validate the output using a Cathode Ray Oscilloscope (CRO)

1.4 Thesis Organization

This thesis is organized into four chapters. Chapter 1 gives an overall introduction to the work, covering the motivation, scope and objectives of the paper. Chapter 2 presents the theoretical background and literature review related to signal generation and different modulation techniques and also highlights the research gap. Chapter 3 discusses the proposed system architecture, design methodology, simulation results, hardware implementation and detailed analysis of the outcomes. Finally, Chapter 4 concludes the work by summarizing the key findings and suggesting possible directions for future enhancements.

1.5 Summary

This chapter presented the motivation, scope of research, and objectives of the proposed work. The need for a flexible and reconfigurable signal generation system using FPGA has been discussed along with the limitations of conventional signal generators. The chapter also provided an overview of the thesis organization. This forms the basis for the detailed study of modulation techniques and system design presented in the subsequent chapters.

Chapter 2

Theoretical Background

2.1 Introduction

The generation of signals is a critical part of modern electronic and communication systems. It is widely used in applications like wireless communications, radar systems, testing equipment and academic labs. In simple terms, generating accurate and controllable signals helps engineers test systems, verify designs, and understand how different communication techniques work. Both RF and baseband signals are important in this process, as they form the basic building blocks of most communication systems. In earlier days, signal generation was mainly done using analog circuits or dedicated signal generators. While these systems are reliable, they are not very flexible. Most of them are designed for specific purposes and cannot be easily modified to generate different types of signals. Also, advanced commercial signal generators are often expensive, which makes them less suitable for student projects and research work. As communication systems continue to evolve, there is a growing need for signal generators that can support multiple modulation techniques and allow easy control of parameters like frequency, amplitude, and phase.

This is where FPGA technology becomes very useful. FPGAs offer a flexible and reconfigurable platform where different signal generation techniques can be implemented using digital design. Unlike traditional hardware, FPGAs can be reprogrammed multiple times without changing the physical circuit. This makes them highly suitable for developing systems that require multiple functionalities within a single platform. In addition, FPGAs provide high-speed operation and precise timing control, which are important for generating accurate signals. Among the various signal processing techniques, modulation plays a key role in communication systems. Methods like Amplitude Modulation, Frequency Modulation and Phase Modulation are widely used to convey information by varying different parameters of a carrier signal. The importance of each technique depends on the application.

Usually, implementing all these modulation techniques requires separate hardware blocks, which increases system complexity and cost. This is not efficient, especially in academic and research environments. Therefore, it is better to have a single system that can generate multiple modulation schemes without requiring hardware changes. This idea forms the main motivation for this project.

In this work, a reconfigurable multi-modulation signal generator is designed using FPGA technology. The system is based on Direct Digital Synthesis (DDS), which allows precise control over signal parameters such as frequency, amplitude, and phase. By using DDS, different types of modulated signals can be generated digitally with good accuracy. The design is developed using Verilog and implemented using Xilinx Vivado tools.

The overall system is built using a modular approach, where different blocks are designed for waveform generation, modulation, and control. This makes the design easy to understand, modify, and extend. The generated signals are verified through simulation and also tested using hardware by observing the output on a Cathode Ray Oscilloscope (CRO). The results show that the system is able to generate AM, FM, PM, and LFM signals correctly.

The main aim of this work is to show that a single FPGA-based platform can be used to generate multiple modulation signals in a flexible and cost-effective way. This type of system can be useful for students, researchers, and engineers working in communication and signal processing areas. It can also be extended in the future to support more advanced modulation techniques and real-time applications.

2.2 Previous Work

Early research in FPGA-based modulation and waveform generation established the feasibility of implementing digital modulation techniques using reconfigurable hardware. Seelam *et al.* [1] demonstrated FPGA implementation of a satellite telemetry transmitter employing PCM/PSK/PM using Direct Digital Synthesis (DDS), highlighting the flexibility of FPGA-based modulation architectures.

Several researchers have focused on linear frequency modulation (LFM) and radar waveform synthesis. Chen *et al.* [2] proposed a real-time wideband waveform generator for airborne synthetic

aperture radar (SAR) applications, emphasizing high-speed processing and resource optimization. In related work, Chen *et al.* [3] introduced a real-time LFM signal generator based on high-level synthesis techniques for airborne SAR applications. Al-Dujaili and Al-Awadi [4] extended conventional chirp generation by introducing Chirplet signal design using FPGA for improved signal characteristics. Said *et al.* [5] developed an efficient FPGA architecture for LFM pulse compression in pulsed radar systems, improving range resolution performance. Similarly, Yang *et al.* [6] implemented a DDS-based chirp signal generator on FPGA for compact SAR applications.

Research in frequency modulation and communication-oriented signal generation has also contributed significantly. Wung *et al.* [7] investigated FPGA implementation of the HD Radio FM baseband transceiver. Li [8] presented a DDS-based FPGA signal generator architecture for flexible waveform synthesis. Aboutabikh *et al.* [9] developed a digital LFM synthesizer targeting FPGA implementation. Büyükkağıcı and Çiçek [10] proposed a real-time FPGA-based wideband signal generator for radar systems, while Setiawan *et al.* [11] focused on DDS-based signal generation for radar applications. Singh *et al.* [12] provided a comprehensive review of radar systems and DDS-based signal generation techniques.

Further developments have emphasized high-speed processing and multi-waveform generation. Hu *et al.* [13] addressed FPGA-based high-speed signal processing for communication systems. Lei *et al.* [14] proposed a multi-modal FPGA signal generator capable of generating multiple waveform types. Hao [15] presented a portable DDS-based signal generation device, demonstrating compact implementation possibilities.

Additional studies explored advanced modulation and communication schemes. Ipanov *et al.* [16] investigated phase-coded signals combined with frequency modulation. Zhu *et al.* [17] presented FPGA implementation of millimeter-wave transmitter architectures. Isik *et al.* [18] extended FPGA applications toward OTFS modulation for next-generation communication systems.

Broader FPGA research has further supported these developments. Ruiz-Rosero *et al.* [19] presented a scientometric review of FPGA applications. Recent advances in FPGA-based digital signal processing architectures were discussed in [20]. Multi-waveform FPGA signal generation for communication systems was explored in [21], while reconfigurable DDS-based signal generators were presented in [22]. FPGA implementations of digital modulation for software-defined radio

applications were reported in [23]. High-resolution chirp signal generation using DDS and FPGA was investigated in [24].

Although substantial work has been reported on individual modulation schemes, radar waveform synthesis, and FPGA-based signal generation, limited research addresses a unified reconfigurable FPGA platform capable of generating multiple modulation formats such as AM, FM, PM, and LFM within a single architecture. This gap motivates the present work.

2.3 Research Gap

From the literature review, it is observed that considerable research has been carried out in FPGA-based signal generation and modulation techniques. Many studies focus on implementing individual modulation schemes, radar waveform synthesis, and DDS-based signal generation using FPGA platforms. These works demonstrate the advantages of digital approaches in terms of flexibility, accuracy and real-time performance. In addition, FPGA-based implementations have been widely explored for communication systems, radar processing, and high-speed signal applications.

However, most existing works are limited to specific modulation techniques or particular application domains. The majority of reported designs focus on a single waveform or a specialized function and do not provide a unified system capable of generating multiple modulation schemes within a single reconfigurable platform. Furthermore, many implementations emphasize simulation-level validation, while comparatively limited attention is given to complete hardware verification using practical measurement instruments such as a Cathode Ray Oscilloscope (CRO).

Therefore, there is a need for a flexible and reconfigurable FPGA-based system capable of generating multiple modulation techniques such as AM, FM, PM, and LFM within a single architecture, along with both simulation and hardware verification. This work addresses these limitations by developing a multi-modulation signal generator using FPGA and Direct Digital Synthesis (DDS), providing a practical and scalable solution for communication and signal processing applications.

2.4 Modulation Techniques

2.4.1 Amplitude Modulation (AM)

Amplitude Modulation is one of the simplest and most widely used modulation techniques in communication systems. In AM, the amplitude of a high-frequency carrier signal is varied according to the instantaneous value of the message signal, while the frequency and phase of the carrier remain constant. Due to its simple implementation and easy demodulation, AM is commonly used in broadcasting systems due to its simple implementation and ease of demodulation.

In an AM signal, the carrier wave acts as a medium to carry the information present in the message signal. The amplitude of the carrier increases or decreases in proportion to the amplitude of the input message signal. This allows the original information to be embedded within the carrier waveform for transmission over long distances.

The mathematical expression of an amplitude modulated signal is given by:

$$s(t) = A_c[1 + m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

Where:

A_c = amplitude of the carrier signal,

f_c = carrier frequency,

f_m = message signal frequency,

m = modulation index.

The modulation index (m) determines the depth of modulation and plays a crucial role in signal quality. For distortion-free transmission, the value of m should be less than or equal to 1. Although AM is easy to generate and requires relatively simple circuitry, it is highly susceptible to noise and interference since noise primarily affects the amplitude of the signal.

Amplitude Modulation is simple to generate and requires less complex circuitry compared to other modulation techniques. However, it is more susceptible to noise and interference, since noise primarily affects the amplitude of the signal. Despite this limitation, AM is still widely used in applications such as radio broadcasting and basic communication systems.

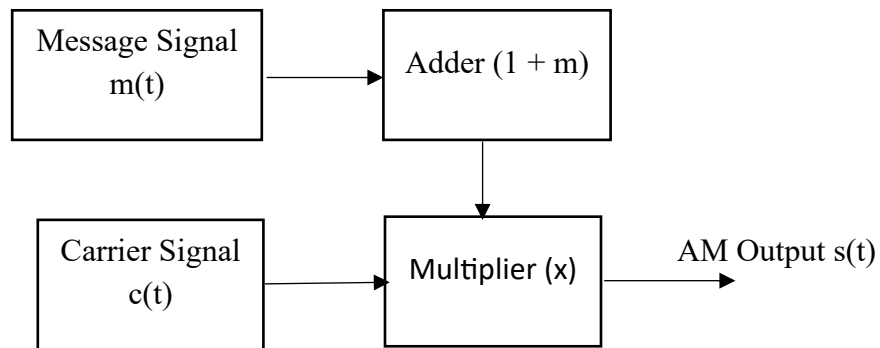


Figure 2.1: Block Diagram of Amplitude Modulation

2.4.2 Frequency Modulation (FM)

Frequency Modulation (FM) is a widely used modulation technique in which the frequency of the carrier signal is varied in accordance with the amplitude of the message signal, while the amplitude of the carrier remains constant. FM is popular in communication systems because it offers better noise immunity compared to amplitude modulation.

In FM, the instantaneous frequency of the carrier increases or decreases depending on the amplitude of the input message signal. This frequency variation allows the information to be effectively encoded and transmitted without affecting the signal strength.

The mathematical expression of an FM signal is given by:

$$s(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$$

Where:

A_c = carrier amplitude,

f_c = carrier frequency,

f_m = message signal frequency,

β = modulation index.

The modulation index (β) represents the maximum frequency deviation and directly influences the bandwidth and noise performance of the signal. A higher modulation index improves noise rejection but requires larger bandwidth. Although FM requires more complex circuitry and occupies wider bandwidth than AM, its superior resistance to noise makes it highly suitable for high-quality radio broadcasting and wireless communication applications.

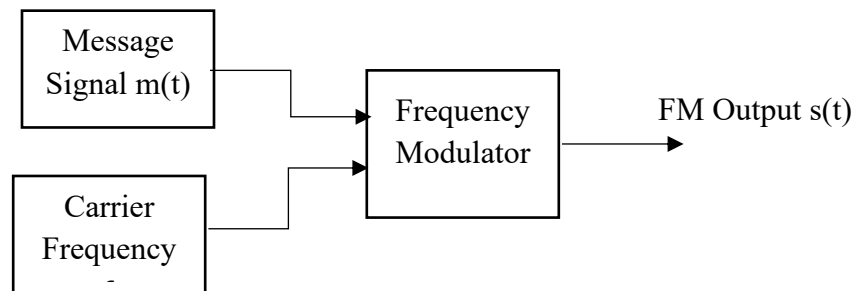


Figure 2.2: Block Diagram of Frequency Modulation

2.4.3 Phase Modulation (PM)

Phase Modulation (PM) is a modulation technique in which the phase of the high-frequency carrier signal is varied according to the instantaneous amplitude of the message signal, while keeping the amplitude and frequency of the carrier constant. PM is widely used in modern digital communication systems due to its good noise performance and efficiency.

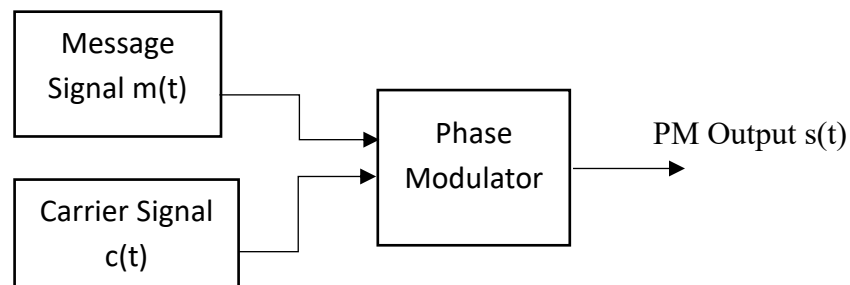


Figure 2.3: Block Diagram of Phase Modulation

In PM, changes in the amplitude of the message signal cause corresponding shifts in the phase of the carrier. This phase variation encodes the information to be transmitted. Since phase and frequency are closely related, a change in phase also results in momentary frequency variations.

The mathematical expression of a phase modulated signal is given by:

$$s(t) = A_c \cos(2\pi f_c t + k_p m(t))$$

Where:

A_c = amplitude of the carrier signal,

f_c = carrier frequency,

k_p = phase sensitivity constant,

$m(t)$ = message signal.

The phase deviation in PM depends on the amplitude of the message signal and the phase sensitivity factor. Unlike frequency modulation, where the frequency varies continuously, PM directly alters the phase of the carrier signal. However, both FM and PM are closely related, as a change in phase also results in a change in frequency.

Compared to AM, PM offers better immunity to amplitude noise. However, it requires more complex circuitry and precise synchronization for accurate demodulation. Phase modulation forms the basis for many digital modulation schemes and is commonly used in advanced communication systems.

2.4.4 Linear Frequency Modulation (LFM)

Linear Frequency Modulation (LFM), commonly known as a chirp signal, is a special type of frequency modulation where the frequency of the carrier signal varies linearly with time. It is extensively used in radar and sensing applications because it provides excellent range resolution and good performance in noisy environments.

In LFM, the instantaneous frequency of the signal increases or decreases at a constant rate over a specific time duration. This gradual frequency sweep allows the signal to cover a wide bandwidth, which is particularly useful for accurate distance and velocity measurement in radar systems.

The mathematical expression of an LFM signal is given by:

$$s(t) = A \cos(2\pi f_0 t + \pi k t^2)$$

Where:

A = amplitude of the signal,

f_0 = initial frequency,

k = chirp rate,

t = time.

The chirp rate (k) determines how rapidly the frequency changes over time. A higher chirp rate results in a wider frequency sweep, which directly improves the resolution of the system. One of the major advantages of LFM is its ability to achieve high range resolution without requiring high peak power. However, it typically requires additional signal processing techniques such as pulse compression at the receiver side.

One of the main advantages of LFM is its ability to achieve high resolution without requiring high peak power. However, it requires additional signal processing techniques for proper analysis and implementation. Despite this, LFM is widely used in radar systems, wireless communication, and signal processing applications.

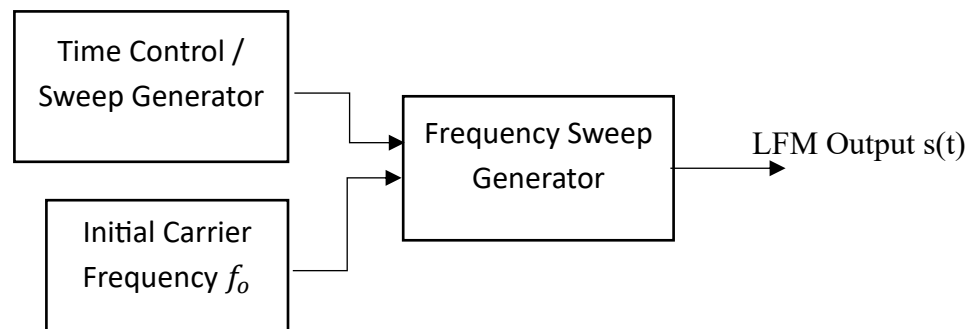


Figure 2.4: Block Diagram of Linear Frequency Modulation

2.4.5 Direct Digital Synthesis (DDS)

Direct Digital Synthesis (DDS) is a popular digital technique used to generate precise and stable waveforms such as sine, cosine, and various modulated signals. It is widely adopted in FPGA-based systems due to its excellent frequency resolution, fast switching speed, and high accuracy.

In DDS, the waveform is generated digitally rather than using traditional analog oscillators. The basic principle involves a phase accumulator that increments at a rate determined by a frequency control word. The output of the phase accumulator is used to address a lookup table (LUT) containing precomputed waveform samples. These samples are then passed through a Digital-to-Analog Converter (DAC) to produce the analog output signal.

By simply changing the frequency control word, the output frequency can be adjusted with high precision without modifying any hardware. This makes DDS highly flexible and suitable for implementing different modulation techniques by dynamically controlling frequency, phase, or amplitude.

DDS offers several advantages over conventional analog signal generation techniques. It provides high frequency resolution, fast switching capability, and excellent stability. It also allows easy implementation of different modulation techniques by digitally controlling phase and frequency parameters. However, DDS may require additional hardware resources such as memory for lookup tables and can introduce quantization errors at very high frequencies.

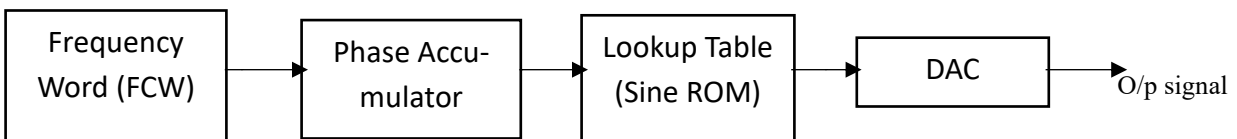


Figure 2.5: Block Diagram of Direct Digital Synthesis

Technique	Parameter Var- ied	Bandwidth	Noise Immunity	Application
AM	Amplitude	Low	Low	Radio Broad- casting
FM	Frequency	High	High	FM Radio
PM	Phase	Moderate	High	Digital Commu- nication
LFM	Frequency (Lin- ear Sweep)	High	Good	Radar Systems

Table 2.1 Comparison of Modulation Techniques

2.5 Summary

This chapter presented the theoretical background and literature review related to signal generation and modulation techniques. Various modulation methods including Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM), and Linear Frequency Modulation (LFM) were discussed along with their basic principles and characteristics. The concept of Direct Digital Synthesis (DDS) was also introduced as an effective technique for digital signal generation. In addition, the research gap was identified based on existing works. These concepts form the foundation for the design and implementation of the proposed FPGA-based multi-modulation signal generator discussed in the next chapter.

Chapter 3

Proposed Work and Methodology

3.1 Introduction

In this chapter the proposed work and methodology for the design of multi modulation signal generator using FPGA technology is presented. The aim of this work is to have a flexible system that can produce different modulations signals such as Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM) and Linear Frequency Modulation (LFM) in a single platform. In conventional systems, separate hardware is often required to generate each modulation technique, which increases system complexity and cost. To overcome this limitation, a unified approach is adopted in this work, where multiple modulation signals are generated using a single FPGA-based architecture.

The proposed system is based on Direct Digital Synthesis (DDS), which provides precise control over signal parameters such as frequency and phase. By using digital design techniques, the system can be easily reconfigured without modifying the hardware. This makes it suitable for applications in communication, signal processing, and testing environments.

The design process begins with verification of modulation signals using MATLAB. After validating the basic waveform characteristics, the system is implemented using Verilog HDL in Xilinx Vivado. The design is then simulated to verify correct operation and finally implemented on FPGA hardware. The output signals are observed using a Cathode Ray Oscilloscope (CRO) to confirm real-time performance.

This chapter describes the overall system design, methodology, and implementation details of the proposed multi-modulation signal generator.

3.2 Multi-Modulation Signal Generation

The proposed system is designed to generate multiple types of modulation signals using a single FPGA-based platform. In this work, the modulation techniques considered are Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM) and Linear Frequency Modulation (LFM). Each of these techniques changes a particular parameter of the carrier signal to represent the in-put information in different forms.

In Amplitude Modulation, the amplitude of the carrier signal is varied according to the input signal, but the frequency and phase of the carrier signal are kept constant. In Frequency Modulation, the amplitude of the carrier is kept constant but the frequency of the carrier is varied in accordance with the input signal. In Phase Modulation, the phase of the carrier signal is varied in accordance with the input signal. In Linear Frequency Modulation, the frequency changes linearly with time, producing a chirp signal that is widely used in radar and sensing systems.

In this work, all modulation techniques are implemented using a digital approach based on Direct Digital Synthesis (DDS). The DDS module serves as the core signal generator, producing a stable and high-resolution carrier signal. By controlling the phase increment and phase offset values, different modulation effects can be achieved efficiently within the same system. The modulation process is carried out using arithmetic and control logic blocks such as adders, multipliers, and counters. These components modify the DDS-generated signal according to the required modulation type. Instead of implementing separate hardware for each modulation technique, a unified structure is used, where common resources are shared among different operations.

A selection control signal is used to choose the desired modulation output. Based on this control input, the corresponding modulation logic is activated, and the output is routed to the final stage. This allows dynamic switching between AM, FM, PM, and LFM without requiring any hardware modification.

The proposed approach reduces hardware complexity and improves resource utilization in the FPGA. It also enhances flexibility, making the system suitable for real-time signal generation applications where multiple modulation schemes are required within a single platform.

3.3 Overall System Architecture

The overall architecture of the proposed multi-modulation signal generator is shown in Figure 3.1. The system is designed using FPGA and implements multiple modulation techniques within a single integrated framework. The architecture consists of several functional blocks including clock generation, phase generation, DDS module, modulation logic, signal selection unit, and output interface. The design is structured in a modular manner to allow efficient implementation and easy reconfiguration.

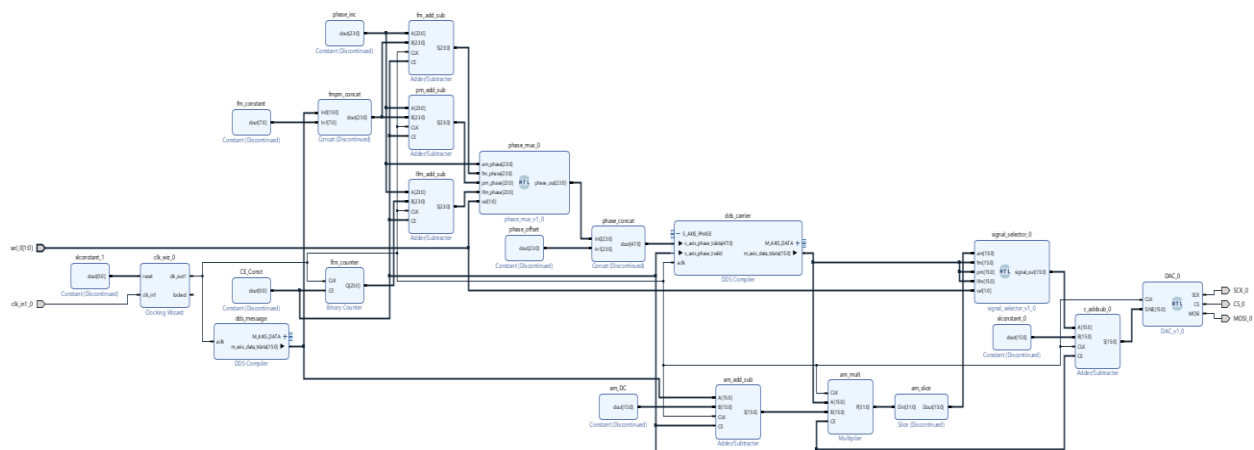


Figure 3.1: Optimized Multi-Modulation Signal Generator Architecture

The system begins with a clocking module, where the input clock is processed using a clock wizard to generate a stable clock signal required for synchronous operation. This clock drives all the internal blocks of the system.

The DDS module forms the core of the architecture and is responsible for generating the carrier signal. The DDS operates based on phase input values, which are generated and modified using phase control logic. Blocks such as phase increment, adders, and concatenation units are used to generate appropriate phase values required for different modulation techniques.

For frequency and phase-based modulations, counters and phase offset blocks are used to dynamically control the frequency and phase variations of the carrier signal. In the case of Linear Frequency Modulation (LFM), a counter-based approach is employed to achieve a linear frequency sweep over time.

For amplitude modulation, arithmetic blocks such as adders and multipliers are used to vary the amplitude of the carrier signal. The AM signal is generated by combining the carrier with a scaling factor and then processed through slicing blocks to obtain the required output format.

A signal selector block is used to choose the desired modulation output based on a selection input. This block routes one of the modulation signals (AM, FM, PM, or LFM) to the output at a given time. This enables dynamic switching between different modulation techniques within the same hardware.

Finally, the selected signal is passed through an output processing stage and fed to a Digital-to-Analog Converter (DAC), which converts the digital signal into an analog waveform. The analog output is then available for observation using external measurement equipment such as a CRO.

The proposed architecture efficiently utilizes shared resources such as the DDS module and arithmetic units, reducing hardware complexity and improving overall system performance.

3.4 Design Methodology

The development of the proposed multi-modulation signal generator follows a structured and sequential design approach, as illustrated in the flowchart. Each stage in the design process plays a significant role in ensuring the correctness, efficiency, and practical implementation of the system.

The design begins with signal analysis, where appropriate modulation techniques are selected based on system requirements and application needs. This step involves identifying suitable modulation schemes such as AM, FM, PM, and LFM, considering factors like signal characteristics, implementation complexity, and hardware feasibility. Following this, existing signal generation techniques are reviewed to understand their working principles and limitations. This analysis helps in identifying the drawbacks of conventional approaches, particularly in terms of flexibility and hardware resource utilization, thereby motivating the need for a unified FPGA-based solution.

In the next stage, the selected modulation techniques are validated using MATLAB. This step is crucial for verifying waveform behavior and ensuring that the generated signals meet theoretical expectations. MATLAB simulation provides a reference model for comparing results obtained from hardware implementation. After validation, the modulation signals are designed using Verilog HDL. At this stage, individual functional modules are developed using digital logic, focusing on accurate representation of modulation behavior through arithmetic and control operations.

The design is then implemented using IP block design in Xilinx Vivado. Various components such as adders, multipliers, counters, and DDS modules are interconnected to form a complete signal generation system. This approach enables efficient hardware realization using pre-designed and optimized blocks. Subsequently, a combined architecture is developed by integrating all modulation techniques into a single unified system. This step focuses on optimizing the design by sharing

common resources, thereby reducing hardware complexity and improving overall system efficiency.

Finally, the complete architecture is implemented on FPGA hardware along with a DAC for signal output. The generated signals are experimentally verified using a Cathode Ray Oscilloscope (CRO) to evaluate real-time performance and validate the correctness of the design.

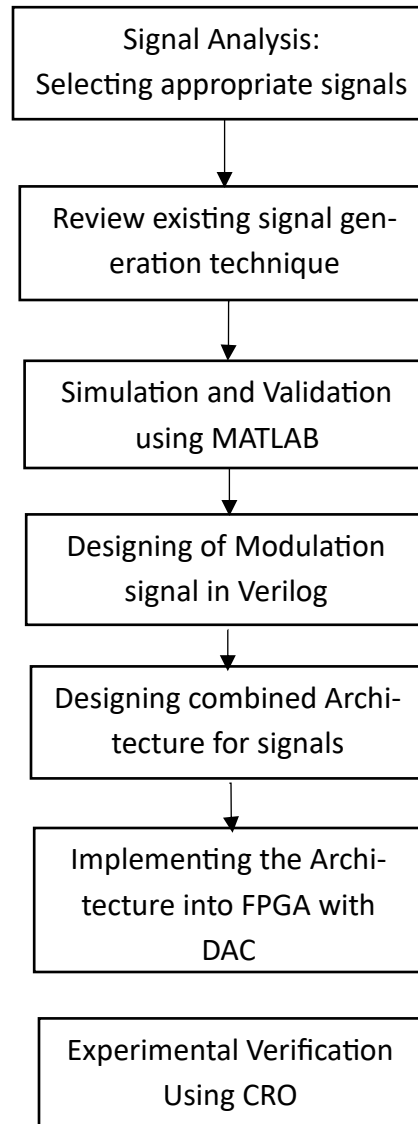


Figure 3.2: Design Flow of the Proposed Multi-Modulation Signal Generator

3.5 Results

3.5.1 MATLAB-Based Analytical Verification

The modulation signals are initially validated using MATLAB to verify their theoretical behavior and waveform characteristics. This stage serves as an analytical reference for evaluating the correctness of the proposed system.

The generated waveforms clearly demonstrate the distinct properties of each modulation technique. The AM signal exhibits variation in amplitude while maintaining a constant carrier frequency. In contrast, the FM signal shows variation in frequency with constant amplitude. The PM signal reflects phase variation of the carrier signal, while maintaining a uniform amplitude profile. The LFM signal demonstrates a linear increase in frequency over time, indicating its chirp-like behavior. The MATLAB results confirm that all modulation schemes are generated in accordance with their theoretical definitions. These results provide a reliable baseline for comparison with simulation and hardware implementation outputs.

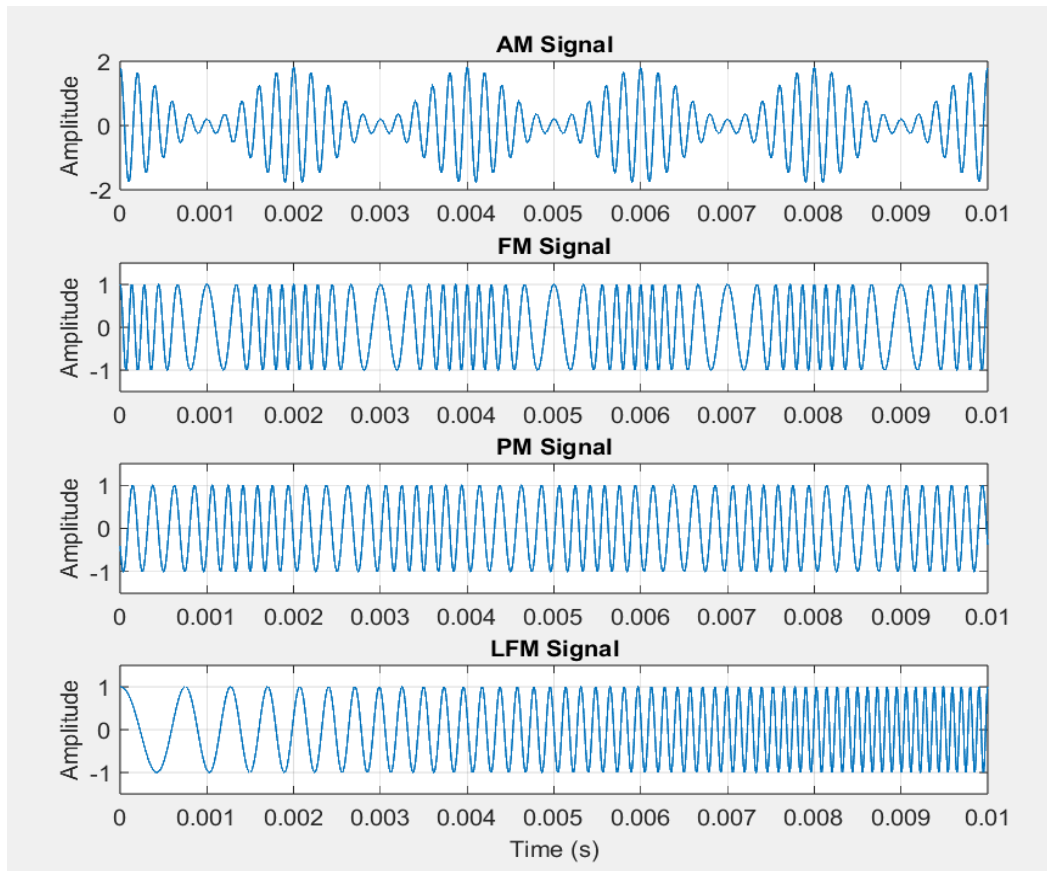


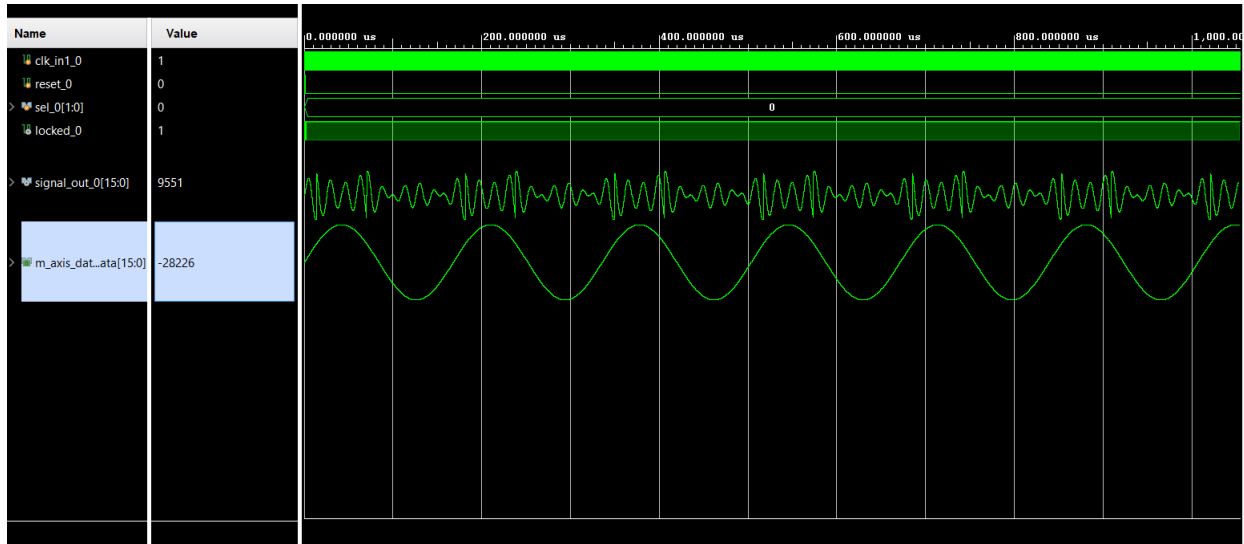
Figure 3.3: MATLAB Generated AM, FM, PM, and LFM Signals

3.5.2 HDL Simulation Results

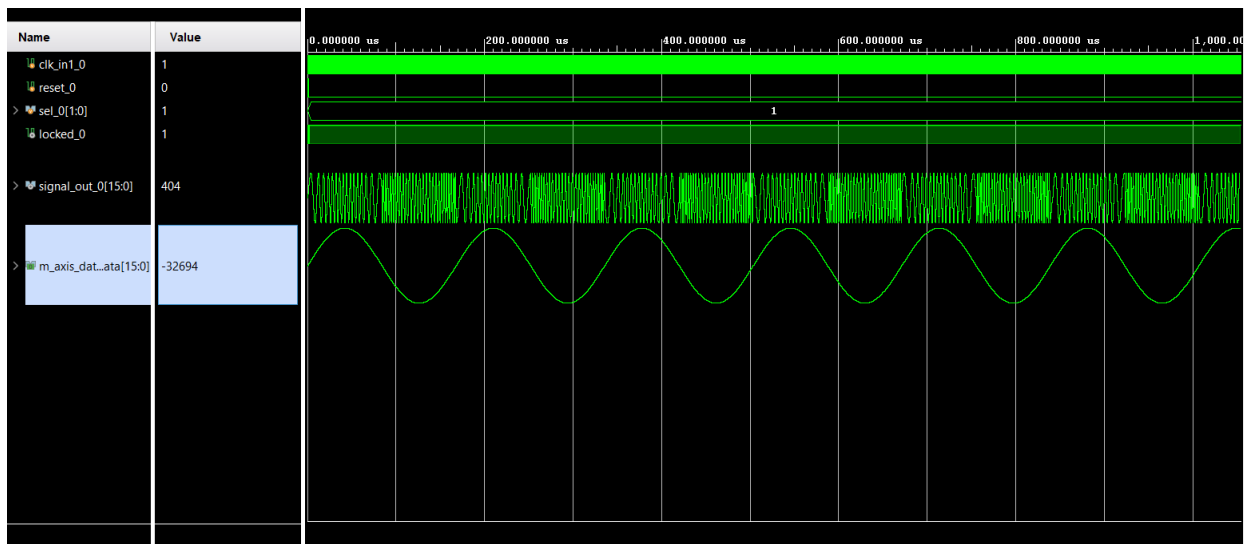
The proposed multi-modulation signal generator is further validated through HDL simulation using Xilinx Vivado. This stage plays a critical role in verifying the functional behavior of the digital architecture prior to hardware implementation. The simulation ensures correct interaction between various modules, including phase generation, arithmetic processing units, and signal selection logic. The simulated waveforms confirm that the system successfully generates all targeted modulation schemes within a unified framework. The AM signal demonstrates controlled envelope variation, indicating correct amplitude scaling of the carrier signal. The FM signal reflects dynamic variation in instantaneous frequency achieved through modulation of the phase increment values in the DDS architecture.

Similarly, the PM signal exhibits phase displacement of the carrier waveform, resulting from direct phase manipulation. Due to the inherent relationship between frequency and phase, FM and PM signals exhibit similar visual characteristics in the time domain, although their modulation principles differ in implementation. The LFM signal demonstrates a continuous and linear increase in frequency over time, confirming the correct realization of chirp signal generation using counter-driven phase control mechanisms. This validates the capability of the proposed system to support advanced modulation schemes within a digital framework.

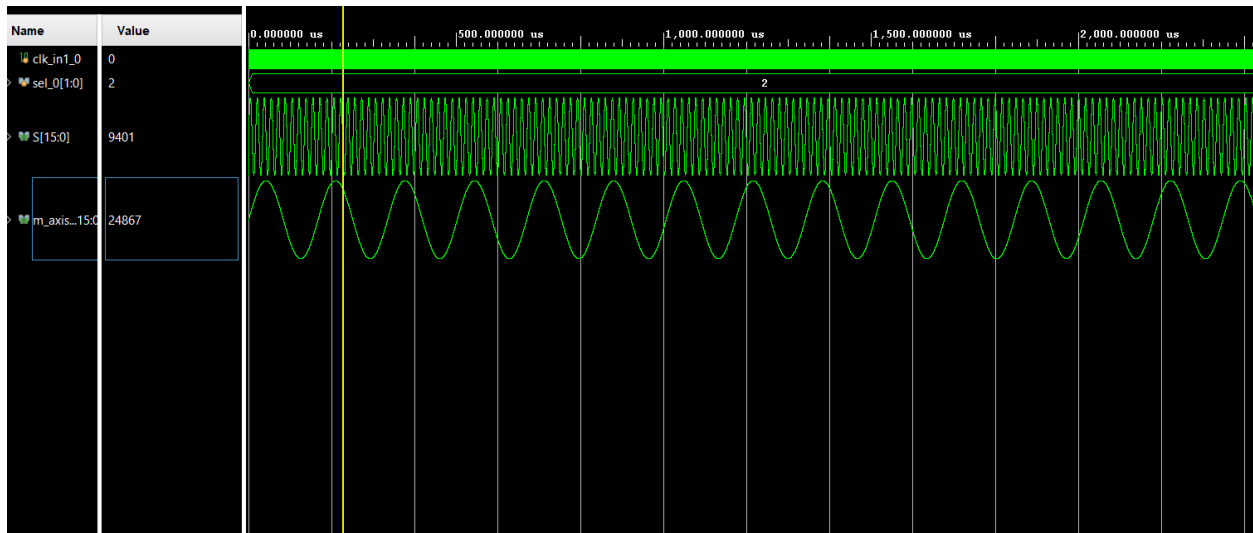
The simulation results establish the functional correctness and stability of the design, ensuring that the system is suitable for reliable hardware implementation.



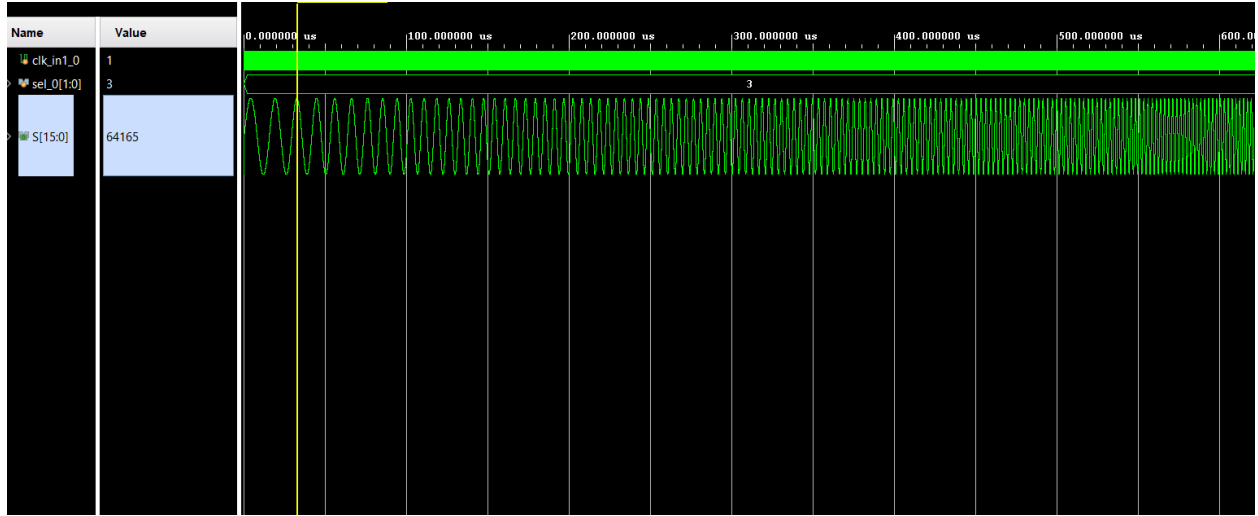
(a) Amplitude Modulation Simulation



(b) Frequency Modulation Simulation



(c) Phase Modulation Simulation



(d) LFM Simulation

Figure 3.4: HDL Simulation Results of (a) AM, (b) FM, (c) PM, and (d) LFM Signals

3.5.3 Experimental Hardware Setup

The experimental validation of the proposed multi-modulation signal generator is carried out using a hardware setup comprising an FPGA development board, a Digital-to-Analog Converter (DAC), and a Cathode Ray Oscilloscope (CRO). This setup enables real-time verification of the digitally generated signals under practical operating conditions.

The FPGA is configured using Xilinx Vivado design tools, where the synthesized design is programmed onto the hardware platform. The digital output signals generated by the FPGA are interfaced with a DAC to convert them into corresponding analog waveforms. These analog signals are then fed into the CRO for visualization and analysis.

The integration of the FPGA with external measurement instruments allows evaluation of key performance aspects such as signal integrity, timing behavior, and waveform accuracy. The use of a CRO provides direct observation of modulation characteristics, thereby facilitating validation of the system beyond simulation.

This experimental arrangement ensures an end-to-end verification framework, bridging the gap between digital design and real-world signal generation.

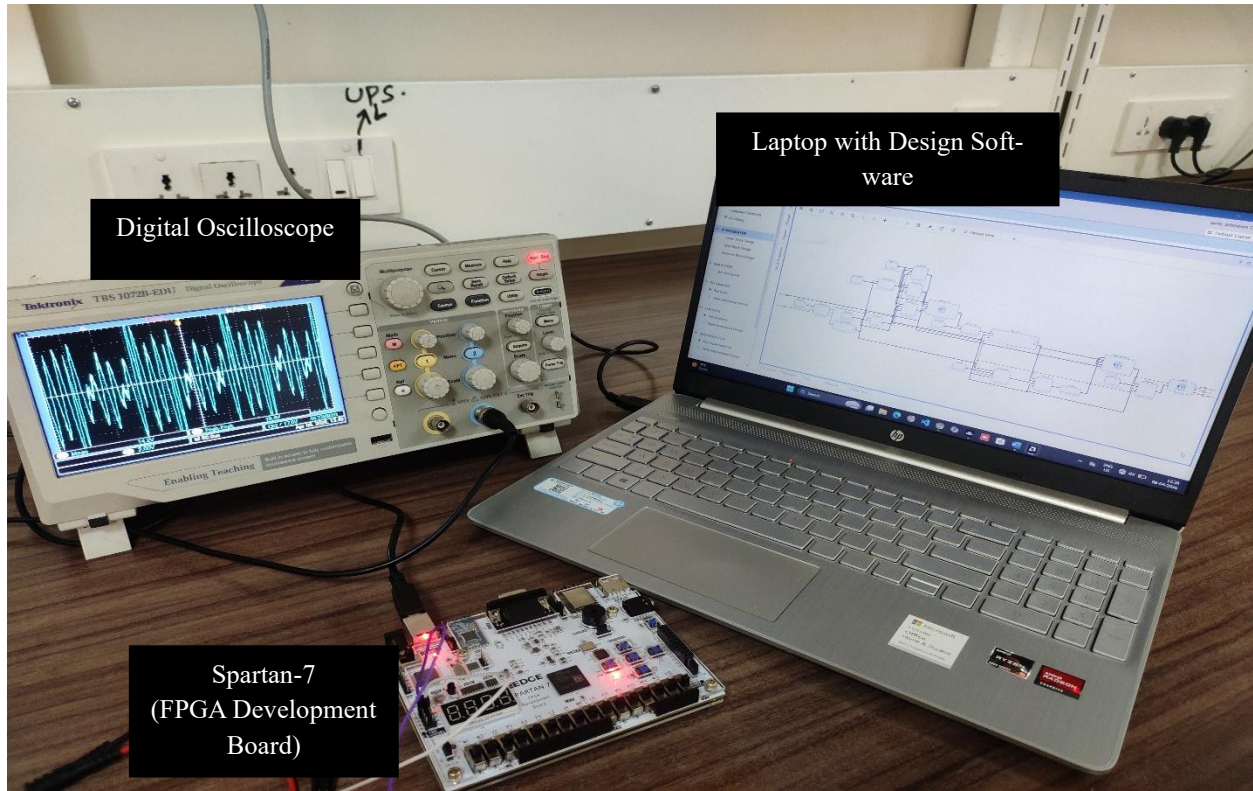


Figure 3.5: Experimental Setup for FPGA-Based Multi-Modulation Signal Generation

The developed setup demonstrates the practical feasibility of the proposed architecture for real-time signal generation applications.

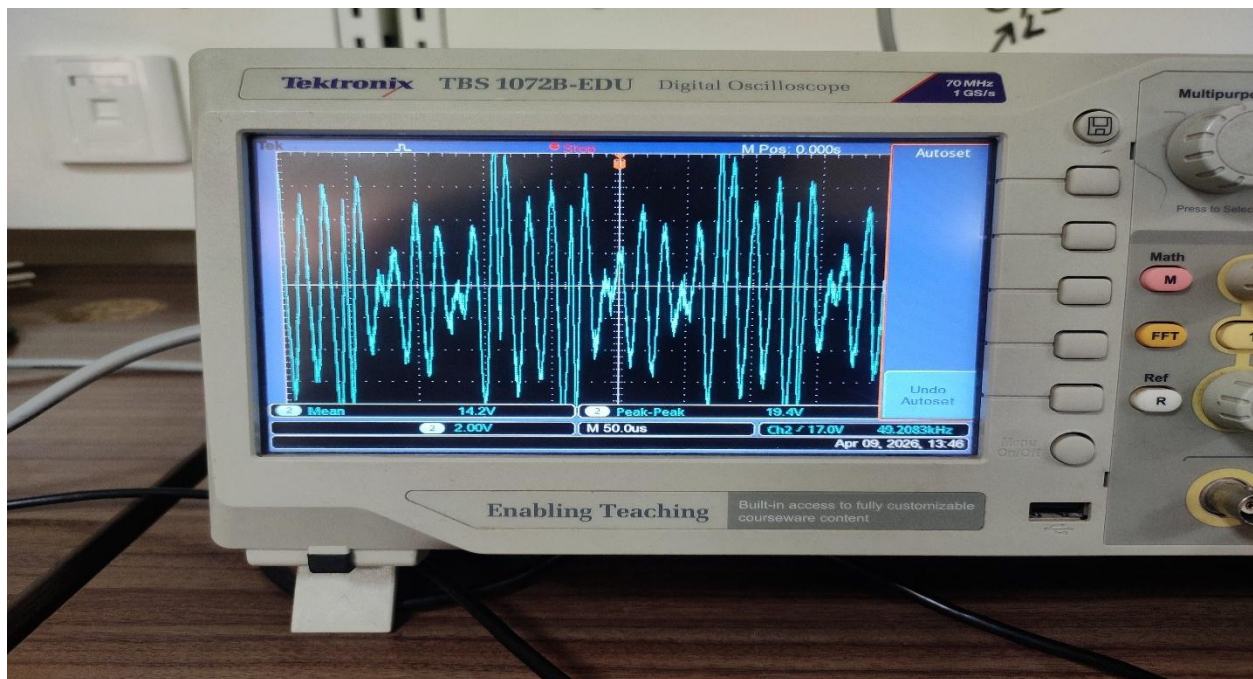
3.5.4 Hardware Verification Using CRO

The final stage of validation involves real-time verification of the generated modulation signals using a Cathode Ray Oscilloscope (CRO). This step is essential for assessing the practical performance of the proposed system, including signal fidelity, timing accuracy, and the impact of hardware constraints. The analog output signals obtained from the DAC are observed on the CRO, where the waveform characteristics of each modulation scheme are analyzed. The AM signal exhibits clear envelope variation corresponding to the modulating signal, confirming correct amplitude modulation. The FM signal demonstrates variation in instantaneous frequency, while maintaining constant amplitude, validating the dynamic phase increment mechanism implemented in the digital domain.

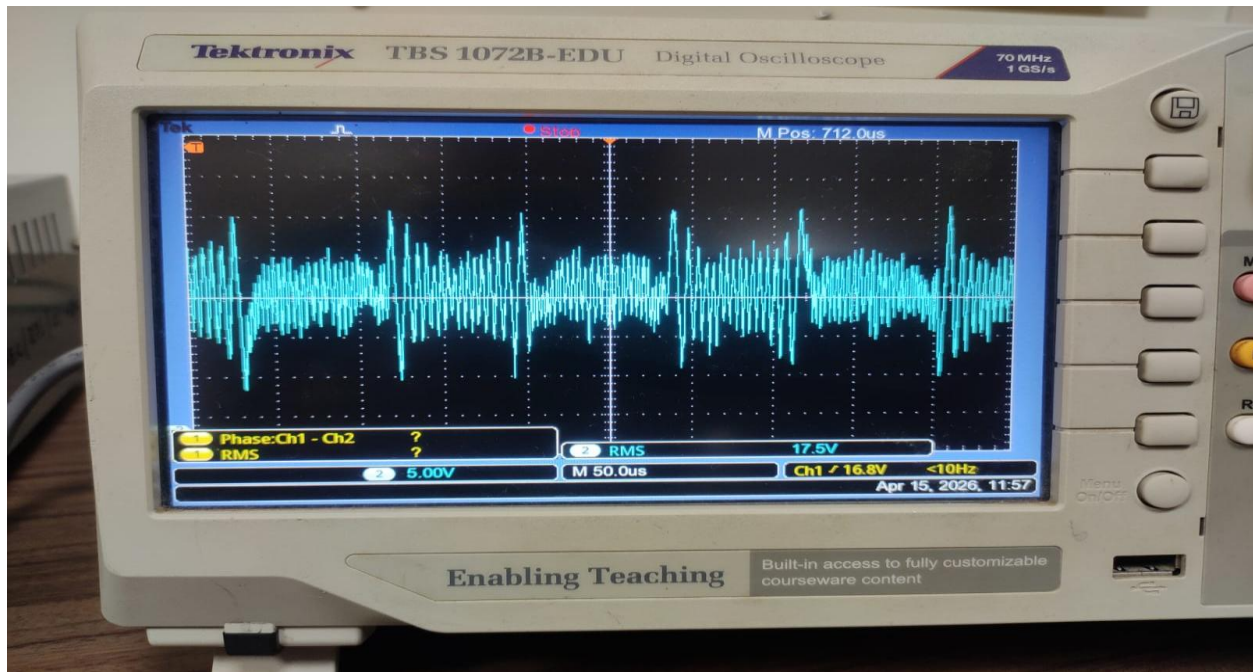
Similarly, the PM signal shows phase variation of the carrier waveform. Due to the intrinsic relationship between frequency and phase, the FM and PM signals appear similar in the time-domain representation, although their modulation processes differ at the implementation level.

The LFM signal, which ideally exhibits a linear frequency sweep, is not distinctly observable in the CRO output. This limitation arises due to practical factors such as finite DAC resolution, sampling constraints, and bandwidth limitations of the measurement setup. These factors affect the visibility of rapid frequency variations in real-time observation.

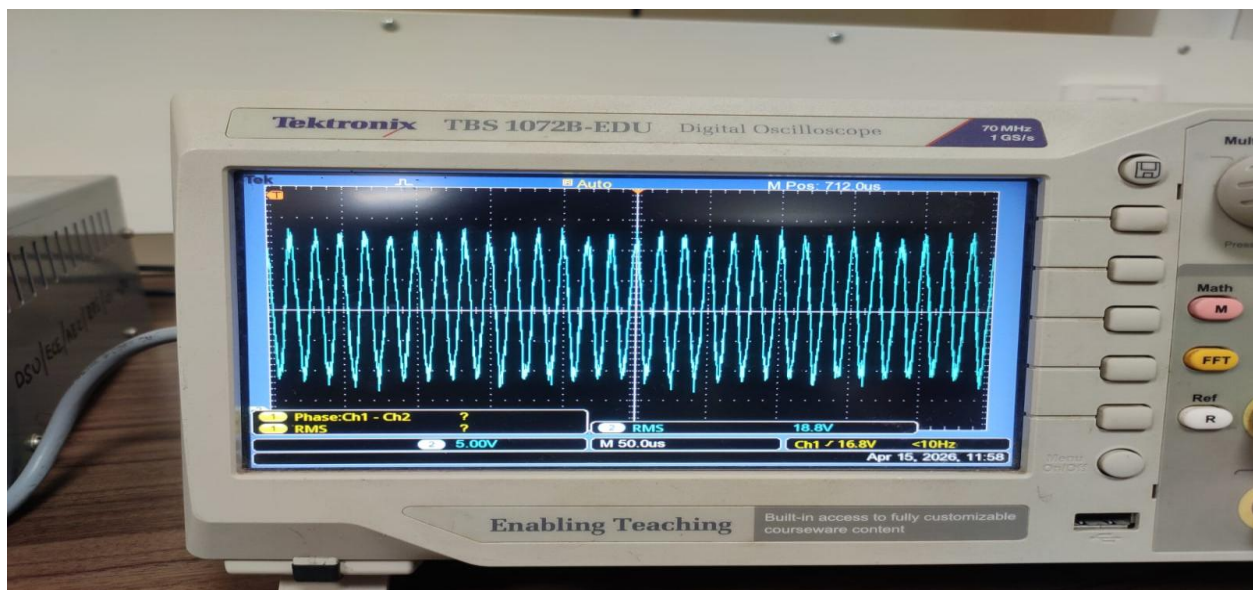
Despite these limitations, the CRO results confirm the successful hardware realization of the proposed multi-modulation signal generator. The observed outputs are consistent with simulation results for AM, FM, and PM signals, thereby validating the effectiveness of the implemented architecture.



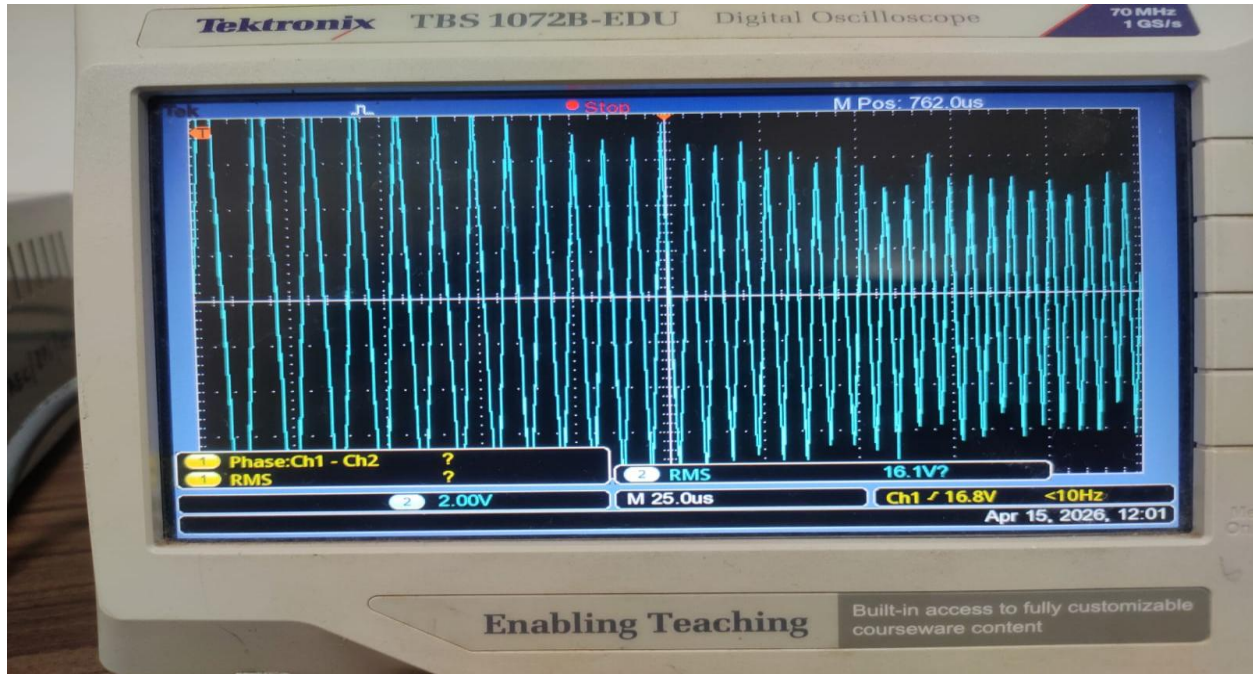
(a) AM Signal



(b) FM Signal



(c) PM Signal



(d) LFM Signal

Figure 3.6: CRO Outputs of (a) AM, (b) FM, (c) PM, and (d) LFM Signals

Modulation	MATLAB Verifica- tion	HDL Simulation	CRO Verification
AM	Verified	Verified	Verified
FM	Verified	Verified	Verified
PM	Verified	Verified	Verified
LFM	Verified	Verified	Limited Visibility

Table 3.1: Simulation and Hardware Verification Summary

3.6 Discussion

The results obtained from MATLAB analysis, HDL simulation, and hardware implementation provide a comprehensive validation of the proposed multi-modulation signal generator. The consistency observed across these stages demonstrates the correctness and robustness of the implemented architecture.

The MATLAB-based analytical results establish the theoretical foundation of the modulation schemes, clearly illustrating the expected behavior of AM, FM, PM, and LFM signals. These

results serve as a reference for evaluating the performance of the digital implementation. The HDL simulation results confirm that the proposed architecture successfully translates theoretical models into a functional digital design. The use of DDS-based phase accumulation and arithmetic processing enables efficient realization of multiple modulation schemes within a unified framework. The observed similarity between FM and PM signals in the time domain is consistent with their inherent dependence on phase variation.

The hardware verification using FPGA and CRO further validates the practical feasibility of the system. The AM, FM, and PM signals are successfully generated and observed with characteristics that closely match simulation results. However, the LFM signal is not distinctly visible in the CRO output due to practical limitations such as DAC resolution, sampling constraints, and bandwidth limitations of the measurement setup. Despite these limitations, the overall system demonstrates reliable performance and flexibility in generating multiple modulation schemes using a single hardware platform. The integration of different modulation techniques within a reconfigurable FPGA architecture highlights the scalability and efficiency of the proposed design.

The minor deviations observed between simulation and hardware results are attributed to real-world non-idealities, which are expected in practical implementations. These factors do not significantly affect the overall functionality of the system.

Chapter 4

Conclusion and Future Scope

4.1 Conclusion

This work presents the design and implementation of a reconfigurable multi-modulation signal generator using FPGA technology based on Direct Digital Synthesis (DDS). The primary objective of the project was to develop a flexible and efficient system capable of generating multiple modulation schemes, namely Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM), and Linear Frequency Modulation (LFM), within a single hardware platform.

A unified architecture has been proposed in which different modulation techniques are implemented using shared digital resources. The use of DDS enables precise control over signal parameters such as frequency and phase, allowing accurate waveform generation. By integrating modulation techniques into a single system, the design reduces hardware complexity and improves resource utilization compared to conventional approaches that require separate modules for each modulation type.

The proposed system is validated through a multi-stage verification process. Initially, MATLAB is used to analyze and verify the theoretical behavior of the modulation signals. The results confirm that the generated waveforms follow expected characteristics such as amplitude variation in AM, frequency deviation in FM, phase variation in PM, and linear frequency sweep in LFM.

The design is then implemented using Verilog HDL and simulated in Xilinx Vivado. The simulation results demonstrate correct functional behavior of the system, confirming the successful implementation of modulation logic, phase control mechanisms, and signal selection architecture. It is observed that FM and PM signals exhibit similar characteristics in the time domain due to their dependence on phase variation, which is consistent with theoretical expectations.

Further validation is carried out through hardware implementation on an FPGA platform. The generated signals are converted into analog form using a DAC and observed using a Cathode Ray Oscilloscope (CRO). The experimental results confirm the successful generation of AM, FM, and PM signals with waveform characteristics closely matching simulation results. However, the LFM

signal is not distinctly observed in the CRO output due to practical limitations such as DAC resolution, sampling constraints, and bandwidth limitations of the measurement setup. Despite this, the correctness of the LFM signal is verified through simulation.

Overall, the proposed system demonstrates a robust and efficient approach for multi-modulation signal generation using FPGA. The ability to generate multiple modulation schemes within a single reconfigurable platform highlights the flexibility and scalability of the design. The integration of DDS-based signal generation with FPGA implementation makes the system suitable for applications in communication systems, signal processing, and academic experimentation.

The results obtained from analytical, simulation, and hardware validation confirm the effectiveness of the proposed architecture, establishing it as a reliable solution for real-time multi-modulation signal generation.

4.2 Future Scope

The proposed FPGA-based multi-modulation signal generator provides a flexible platform for generating multiple modulation schemes. However, there are several areas where the system can be further improved and extended to enhance its performance and applicability.

One of the primary improvements involves the use of higher-resolution Digital-to-Analog Converters (DACs) and high-speed FPGA platforms. This would improve the quality of the generated analog signals and enable better visualization of complex signals such as Linear Frequency Modulation (LFM) in real-time. Enhancing the sampling rate and bandwidth of the system can significantly improve signal accuracy and reduce distortion.

The system can also be extended to support advanced digital modulation techniques such as Quadrature Amplitude Modulation (QAM), Phase Shift Keying (PSK), and Orthogonal Frequency Division Multiplexing (OFDM). Incorporating these modulation schemes would make the system more suitable for modern wireless communication applications and software-defined radio (SDR) systems.

Further optimization of the design can be carried out to reduce hardware resource utilization and power consumption. Techniques such as resource sharing, pipelining, and efficient memory

utilization can be explored to improve the overall performance of the system, especially for embedded and portable applications.

Another potential extension includes the integration of real-time control interfaces, allowing dynamic adjustment of modulation parameters such as frequency, amplitude, and phase. This would enable the system to be used as a programmable signal generator for testing and measurement applications.

In addition, the proposed architecture can be adapted for specialized applications such as radar signal generation, channel modeling, and communication system testing. The LFM signal, in particular, can be further refined and implemented more effectively for radar and sensing applications with improved hardware support.

Overall, the proposed system provides a strong foundation for future research and development in FPGA-based signal generation. With further enhancements, it can be extended into a versatile and high-performance platform for a wide range of communication and signal processing applications.

Abbreviations

AM – Amplitude Modulation

FM – Frequency Modulation

PM – Phase Modulation

LFM – Linear Frequency Modulation

FPGA – Field Programmable Gate Array

DDS – Direct Digital Synthesis

DAC – Digital-to-Analog Converter

CRO – Cathode Ray Oscilloscope

HDL – Hardware Description Language

LUT – Look-Up Table

DSP – Digital Signal Processing

MATLAB – Matrix Laboratory

References

- [1] A. S. B. Seelam, S. B. Prusty, and V. Shahul Hameed, “Design and FPGA implementation of satellite telemetry transmitter employing PCM/PSK/PM using DDS,” in *Proc. IEEE Int. Conf. Electron., Comput., Commun. Control Technol. (ICECCC)*, 2024.
- [2] D. Chen *et al.*, “A novel real-time processing wideband waveform generator of airborne synthetic aperture radar,” *Remote Sens.*, vol. 16, no. 3, pp. 1–18, 2024.
- [3] D. Chen *et al.*, “A novel linear frequency modulation signal generator in real-time processing of airborne SAR,” *J. Eng.*, 2023.
- [4] M. J. Al-Dujaili and A. M. Al-Awadi, “Chirplet signal design by FPGA,” *Int. J. Electr. Comput. Eng.*, vol. 11, no. 3, pp. 2120–2127, 2021.
- [5] H. A. Said, A. E. El-Henawey, and A. A. El-Kouny, “Design and realization of an efficient VLIC architecture for linear frequency modulation pulse compression in pulsed radars using FPGA,” in *Proc. IEEE Conf.*, 2019.
- [6] H. Yang *et al.*, “Implementation of DDS chirp signal generator on FPGA,” in *Proc. IEEE Conf.*, 2019.
- [7] L. Wung *et al.*, “Theoretical design and FPGA implementation of the baseband transceiver for the HD radio FM system,” in *Proc. IEEE Conf.*, 2019.
- [8] S. Li, “Design and implementation of DDS signal generator based on FPGA,” *J. Adv. Sci. Technol.*, 2024.
- [9] K. Aboutabikh *et al.*, “Design and implementation of digital LFM synthesizer using FPGA,” *Int. J. Mod. Res.*, 2024.
- [10] O. Büyükkagıncı and İ. Çiçek, “Real-time FPGA-based wideband signal generator for radar applications,” *IEEE Access*, 2025.
- [11] H. Setiawan *et al.*, “FPGA-based direct digital synthesis signal generation for radar applications,” *IEEE Access*, 2025.

- [12] P. Singh, R. Kumar, and S. Tiwari, "Radar systems and DDS-based signal generation: A review," *J. Electron. Eng.*, 2025.
- [13] S. Hu, Y. Zhang, and X. Liu, "FPGA-based high-speed signal processing for communication systems," *Sensors*, vol. 24, no. 23, 2024.
- [14] H. Lei, X. Wang, and Y. Chen, "Design and simulation of multi-modal signal generator using FPGA," in *Proc. IEEE Int. Conf.*, 2020.
- [15] D. Hao, "DDS-based portable signal generation device," *J. Phys.: Conf. Ser.*, 2021.
- [16] R. N. Ipanov *et al.*, "Phase-code shift keyed signals with frequency modulation," *IEEE Trans.*, 2019.
- [17] W. Zhu, Y. Li, and Z. Wang, "Design of millimeter-wave transmitter based on FPGA," *IEEE Trans.*, 2019.
- [18] M. Isik, A. Sahin, and E. Basar, "FPGA implementation of OTFS modulation for next-generation communication systems," *IEEE*, 2023.
- [19] J. Ruiz-Rosero *et al.*, "FPGA applications: A scientometric review," *Electronics*, vol. 8, no. 8, 2019.
- [20] "Recent advances in FPGA-based digital signal processing architectures," *IEEE Access*, vol. 10, pp. 1–15, 2022.
- [21] "FPGA-based multi-waveform signal generator for communication applications," *IEEE Access*, vol. 11, pp. 1–12, 2023.
- [22] "Design of reconfigurable signal generator using DDS on FPGA," *Springer*, 2022.
- [23] "FPGA-based implementation of digital modulation techniques for SDR applications," *IEEE*, 2021.
- [24] "High-resolution chirp signal generation using DDS and FPGA," *IEEE Trans. Instrum. Meas.*, 2022.

Dissemination of Thesis

The research work presented in this thesis has been disseminated through the following publications and presentations:

Conference Publications

[1] Supreeth K, Vankadara Sai Gayathri, and Dr. Shirshendu Roy, “FPGA-Based Design and Implementation of a Reconfigurable Multi-Modulation Signal Generator Using Direct Digital Synthesis,” planned for submission to IEEE Connect, 2026.

[2] Vankadara Sai Gayathri, Supreeth K and Dr. Shirshendu Roy, “Architectures and algorithms for Sinusoidal Generation in Digital Platforms” Submitted to IEEE Connect, 2026, Paper ID: 51

Technical Reports / Other Contributions

1. Project report titled “FPGA-Based Design and Implementation of a Reconfigurable Multi-Modulation Signal Generator Using Direct Digital Synthesis,” Department of Electronics and Communication Engineering, Dayananda Sagar University, 2026.

FPGA Based Design and Implementation of a Reconfigurable Multi - Modulation Signal Generator using DDS

ORIGINALITY REPORT

3%

SIMILARITY INDEX

3%

INTERNET SOURCES

3%

PUBLICATIONS

1%

STUDENT PAPERS

PRIMARY SOURCES

1

www.wonderslate.com

Internet Source

1%

2

Gaurav Aggarwal, Rishabh Dev Shukla, Innocent Ewean Davidson, Ramesh Bansal. "Integrated Technologies in Electrical, Electronics and Biotechnology Engineering - Proceedings of the 2nd International Conference on Innovative Trends in Electrical, Electronics and Bio-Technology Engineering (ICITEEB 2025), 08th-09th July 2025, Mohali, India", CRC Press, 2026

Publication

1%

3

www.freepatentsonline.com

Internet Source

1%

4

www.satnow.com

Internet Source

1%

5

Submitted to Asia Pacific University College of Technology and Innovation (UCTI)

Student Paper

1%

Exclude quotes On

Exclude matches < 1%

Exclude bibliography On